



# JavaScript



# JavaScript

In this session you will learn

JAVASCRIPT

Different ways  
of writing  
JavaScript

# WHAT IS JAVASCRIPT

- JavaScript is the client-side scripting language of the Web.
- JavaScript is used in millions of Web pages to add functionality, validate forms, detect browsers, and much more.
- JavaScript is the most popular scripting language on the internet, and works in all major browsers, such as Internet Explorer, Firefox, Chrome, etc
- JavaScript is an interpreted language (means that scripts execute without preliminary compilation)
- JavaScript is usually embedded directly into HTML pages

# Are Java and JavaScript the same?

NO!

- Java and JavaScript are two completely different languages in both concept and design!
- Java (developed by Sun Microsystems) is a powerful and much more complex programming language - in the same category as C and C++.

# What can a JavaScript do?



JavaScript gives HTML designers a programming tool



JavaScript can put dynamic text into an HTML page



JavaScript can react to events



JavaScript can read and write HTML elements



JavaScript can be used to validate data



JavaScript can be used to detect the visitor's browser



JavaScript can be used to create cookies

# Where to Put the JavaScript

# 1. Scripts in <head>

Scripts to be executed when they are called, or when an event is triggered, are placed in functions in the head section

## Example

```
<html>
<head>
<script type="text/javascript">
function message()
{
alert("This alert box was called with the onload event");
}
</script>
</head>
<body onload="message()">
</body>
</html>
```

## 2. Scripts in <body>

If you don't want your script to be placed inside a function, or if your script should write page content, it should be placed in the body section.

### Example

```
<html>
<head>
</head>
<body>
<script type="text/javascript">
document.write("This message is written by JavaScript");
</script>
</body>
</html>
```



### 3. Scripts in <head> and <body>

you can have scripts in both the body and the head section.

Example

```
<html>
<head>
<script type="text/javascript">
function message()
{
alert("This alert box was called with the onload event");
}
</script>
</head>
<body onload="message()">
<script type="text/javascript">
document.write("This message is written by JavaScript");
</script>
</body>
</html>
```

## 4. use External JavaScript file

- If you want to run the same JavaScript on several pages, without having to write the same script on every page, you can write a JavaScript in an external file.
- To use the external script, point to the .js file in the "src" attribute of the <script> tag
- Save the external JavaScript file with a .js file extension.

E.G – to create an external JavaScript file that prints Hello Javatpoint in a alert dialog box on click of a button

Index.html

```
<html>
<head>
<script type="text/javascript" src="message.js">
</script>
</head>
<body>
<p>Welcome to JavaScript</p>
<form>
<input type="button" value="click" onclick="msg()"/>
</form>
</body>
</html>
```

message.js

```
function msg(){
  alert("Hello Java");
}
```



# Viva Questions

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- WHAT IS JAVASCRIPT?
- Are Java and JavaScript the same?
- What can we do with JavaScript ?
- What are different ways of writing JavaScript?
- How to include JavaScript in <head> section?
- How to include JavaScript in <body> section?
- Can we write JavaScript in <head> and <body> section?
- Can we use external JavaScript file?
- What is the extension of external JavaScript File?